

Improving the Training of the Projected GAN with Limited Data via Adaptive Feature Augmentation

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Abstract

Training Generative Adversarial Networks (GANs) with limited data has become more and more popular in the past few years. Recently, many studies have applied data augmentation (DA) to transform both real and fake images in image space to improve GANs training with limited data. Among these methods, Projected GAN is the first study that applies the pre-trained network to improve GANs training in feature space. As a result, Projected GAN can better utilize deeper layers of the pre-trained network, thus resulting in impressive results when training with limited data. Despite the success of the Projected GAN training in feature space, feature augmentation is another plausible, complementary, yet not well-researched strategy. In this paper, we propose a simple yet effective feature augmentation for the Projected GAN, called Adaptive Feature Augmentation (AFA), to further improve the training of the Projected GAN with limited data. Specifically, we apply the forward diffusion process to the projected features obtained from both real and fake images in order to augment projected features themselves adaptively, which can effectively provide more overlapping parts between the supports of real and fake projected features distributions during training, thus leading to better performance. Extensive experiments on several low-shot datasets demonstrate that AFA can further improve the training of the Projected GAN and achieve state-of-the-art results compared with other methods.

1. Introduction

In recent years, Generative Adversarial Networks (GANs) [10] have been remarkably successful [17, 20, 21] in generating high-quality data when working with large datasets. However, collecting and cleaning these large datasets can be expensive, time-consuming, and even infeasible. Consequently, training GANs with limited data has received great attention.

Data augmentation (DA) has recently shown its impor-

tance in training GANs with limited data. Many studies [5, 19, 48] apply DA to transform both real and fake images for discriminator (D) and generator (G) to improve GANs training with limited data. Among these methods, Projected GAN [36] has been shown to be the first method that utilizes the pre-trained network, i.e., EfficientNet-lite1 [39], to optimize GANs in feature space. Specifically, Projected GAN projects both real and fake images into a fixed, pre-trained feature space, enabling better utilization of deeper layers in the pre-trained network, leading to impressive results when training with limited data.

Despite the success of the Projected GAN in feature space, feature augmentation is another plausible, complementary, yet not well-researched strategy. Thus, we propose a simple yet effective feature augmentation for the Projected GAN, called Adaptive Feature Augmentation (AFA), to further improve the training of the Projected GAN with limited data. The form of the proposed AFA is motivated by the existing feature space data augmentation approaches [8, 23]. Specifically, AFA utilizes the forward diffusion process [14] to the projected features obtained from both real and fake images using the pre-trained network to augment projected features themselves in the Projected GAN. Then, according to the theory developed in [1, 42], i.e., more overlapping parts between the supports of the optimization distributions in GANs can improve the training of GANs, we prove that AFA can effectively provide more overlapping parts between the supports of real and fake projected features distributions during training (see Theorem 1 in section 3.2), as shown in Figure 1. Furthermore, based on the conclusion in ADA [19], i.e., the loss of sufficient overlap between distributions is a strong indication of overfitting, more overlapping parts between the supports of real and fake projected features distributions produced by AFA can also alleviate the overfitting of D problem [16, 19, 40, 48]. Thus, AFA can improve the training of Projected GAN with limited data and result in better performance.

To sum up, the main contributions of this paper are as follows:

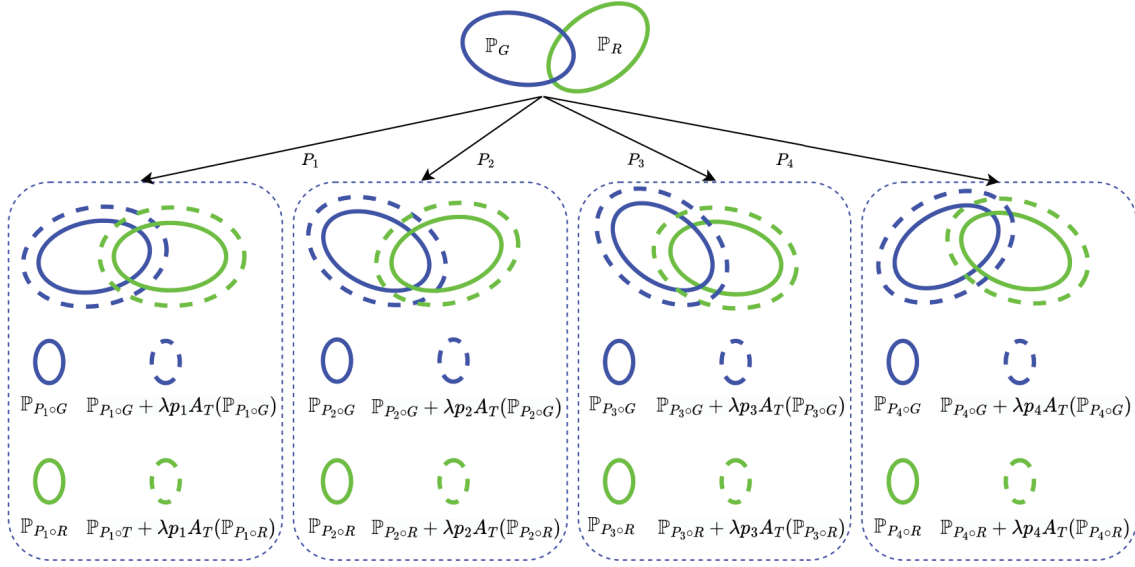


Figure 1. Schematic overview illustrating how AFA can benefit the training of the Projected GAN. Following the Projected GAN [36], we obtain features from four layers of a pre-trained feature network at different resolutions. $\{P_i\}$ is the set of feature projectors which map real and generated images to the discriminator’s input space. $\mathbb{P}_G$ and $\mathbb{P}_R$ represent real and fake images distributions, respectively. $\mathbb{P}_{P_i \circ R}$ and $\mathbb{P}_{P_i \circ G}$ represent the real and fake projected features distributions, respectively. A_T is the forward diffusion process to augment features. λ and p_i are the parameters in AFA defined in section 3.1. Applying AFA for projected features can effectively provide more overlapping parts between the supports of real and fake projected features distributions during training, thus resulting in better performance [1, 42].

1. We propose a novel Adaptive Feature Augmentation (AFA) for the Projected GAN. To our best knowledge, the proposed AFA is the first DA method that acts in feature space to improve the GANs training. Specifically, AFA utilizes the forward diffusion process to augment the projected features obtained from both real and fake images in the Projected GAN. In this case, AFA can effectively provide more overlapping parts between the supports of the real and fake projected features distributions during training, thus leading to better performance.
2. We provide the theoretical analysis of Adaptive Feature Augmentation (AFA) for projected features in the Projected GAN, proving its convergence and rationality. Furthermore, we prove that applying AFA in feature space can provide more overlapping parts between the supports of the real and fake projected feature distributions.
3. Extensive experiments on seven commonly used datasets demonstrate that the proposed AFA can further improve the training of the Projected GAN and achieve state-of-the-art performance compared with other methods. Additionally, AFA only increases the negligible computational cost.

2. Preliminaries

2.1. Generative Adversarial Networks (GANs)

Generative adversarial networks (GANs) [10] is a form of generative models [31, 41] in which a game is played between two players: A generator (G) and a discriminator (D). Specifically, G aims to produce realistic-looking samples with some given noise z to deceive D , while D aims to distinguish whether the input sample is from the generator’s output or real data. The objective function of GANs can be formulated as follows:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim \mathbb{P}_x} [\log D(x)] + \mathbb{E}_{z \sim \mathbb{P}_z} [\log(1 - D(G(z)))]. \quad (1)$$

The parameters of G and D are updated iteratively with gradient descent methods. GANs are known to suffer from training instability, yielding the poor quality and diversity of generated images. To stabilize GANs training as well as improve the quality and diversity of generated images, various approaches have been proposed, focusing on more sophisticated network architectures [4, 29, 30, 32, 45, 46], more stable objective functions [2, 11, 12, 28, 35], and better training strategies [7, 17, 24, 26, 27, 43, 47] to achieve photorealistic results.

2.2. Training GANs with Limited Data

Training GANs with limited data using data augmentation. Recently, data augmentation (DA) has played an important role in improving the performance of training GANs with limited data. Many studies [5, 19, 40, 48] apply DA to both real and fake images for D and G to guide the discriminator avoiding overfitting. The most popular methods are Diff-Augment [48], ADA [19], and Diffusion-GAN [42]. Diff-Augment applies the DAs to both real and fake images for the D and the G without manipulating the target distribution. ADA is similar to Diff-Augment, while it further devises an adaptive approach that controls the strength of data augmentations. Diffusion-GAN applies the forward diffusion process [14] as DA to both real and fake images.

Training GANs with limited data using the pre-trained network. Recently, the pre-trained network has shown its significance in improving the training of GANs with limited data. Many approaches [22, 36] apply the pre-trained network to the discriminator for extracting the features from both real and fake images to improve the training of GANs with limited data. One of the pioneering methods is the Projected GAN. Specifically, Projected GAN applies the pre-trained network, i.e., EfficientNet-lite1 [39], to both real and fake images to extract projected features in image space. Then, Projected GAN applies a multi-scale discriminator architecture which can better utilize deeper layers of the pre-trained network. As a result, Projected GAN can effectively utilize the obtained projected features to benefit the training of GANs with limited data. The loss function of the Projected GAN can be formulated as:

$$\min_G \max_{\{D_l\}} \sum_{l \in \mathcal{L}} (\mathbb{E}_{x \sim \mathbb{P}_x} [\log D_l(P_l(x))] + \mathbb{E}_{z \sim \mathbb{P}_z} [\log(1 - D_l(P_l(G(z))))]), \quad (2)$$

where $\{P_l\}$ is the set of feature projectors which map real and generated images to the discriminator's input space, $\{D_l\}$ is a set of independent discriminators operating on different feature projections and $\mathcal{L} = \{1, \dots, n\}$ is the set to number different projectors.

2.3. Diffusion-based Generative Model

In recent years, Diffusion-based generative models [9, 14, 18] have shown their superiority in image-generation tasks. Diffusion-based generative models assume that $p_\theta(x_0) := \int p_\theta(x_{0:T}) dx_{1:T}$, where $x_1, \dots, x_T$ are latent variables of the same dimensionality as the data $x_0 \sim p(x_0)$. There is a forward diffusion chain that gradually adds noise to the data $x_0 \sim q(x_0)$ with pre-defined variance schedule β_t and variance σ^2 in T steps:

$$\begin{aligned} q(x_{1:T}|x_0) &:= \prod_{t=1}^T q(x_t|x_{t-1}), \\ q(x_t|x_{t-1}) &:= N(x_t; \sqrt{1 - \beta_t}, \beta_t \sigma^2 I). \end{aligned} \quad (3)$$

A notable property is that x_t at an arbitrary time-step t can be sampled in closed form as:

$$q(x_t|x_0) := N(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\sigma^2 I), \quad (4)$$

where $\alpha_t := 1 - \beta_t$, $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$. A variational lower bound [3] is then used to optimize the reverse diffusion chain as:

$$p_\theta(x_{0:T}) := N(x_T; 0, \sigma^2 I) \prod_{t=1}^T p_\theta(x_{t-1}|x_t).$$

2.4. Data Augmentation in Feature Space

In the past few years, many studies [8, 23] have focused on how to apply data augmentation (DA) in the feature space; specifically, they add random Gaussian distribution noise in feature space to achieve better performance. This technique can be formulated as follows:

$$c' = c + \lambda X, \quad X \sim N\{0, \sigma^2\}, \quad (5)$$

where λ is the parameter to scale the noise, c is the original feature and c' is the obtained augmented features.

3. Methodology

3.1. Adaptive Feature Augmentation

The overview of Adaptive Feature Augmentation (AFA) in the Projected GAN is shown in Figure 2. AFA utilizes the forward diffusion process to the projected features obtained from both real and fake images by a pre-trained network to augment projected features themselves adaptively. Following the Projected GAN, we obtain features from four layers L_l of a pre-trained feature network F at different resolutions, i.e., $L_1 = 64^2$, $L_2 = 32^2$, $L_3 = 16^2$, $L_4 = 8^2$. Specifically, all features before Cross-Channel Mixing (CCM) and Cross-Scale Mixing (CSM) processed in the Projected GAN are augmented with AFA to improve the training of the Projected GAN. As a result, AFA can provide more overlapping parts between the supports of real and fake projected features distributions.

As stated in ADA [19], it is well known that ‘a noise augmentation leads to noisy results, even if there is none in the dataset’. The forward diffusion process is also a form of noise augmentation which could cause the leaking of augmentations during training. To alleviate this, we apply a hyperparameter λ to balance the AFA and perform the AFA based on a form of probability, $p_l \in [0, 1]$ for each scale of features. It is intuitive to let p_l be adjusted adaptively based on the degree of overfitting without manual tuning regardless of data scales and properties. To achieve this goal, following the ADA [19] and APA [16], we apply an overfitting heuristic η_l that quantifies the degree of D_l 's overfitting as follows:

$$\eta_l = \mathbb{E}(\text{sign}(D_{l_{real}})), \quad D_{l_{real}} = \text{logit}(D_l(P_l(x))), \quad (6)$$

where $\{P_l\}$ is the set of feature projectors that map real and generated images to the discriminator's input space, $\{D_l\}$ is a set of independent discriminators operating on different feature projections, $\text{sign}()$ indicates the sign function that

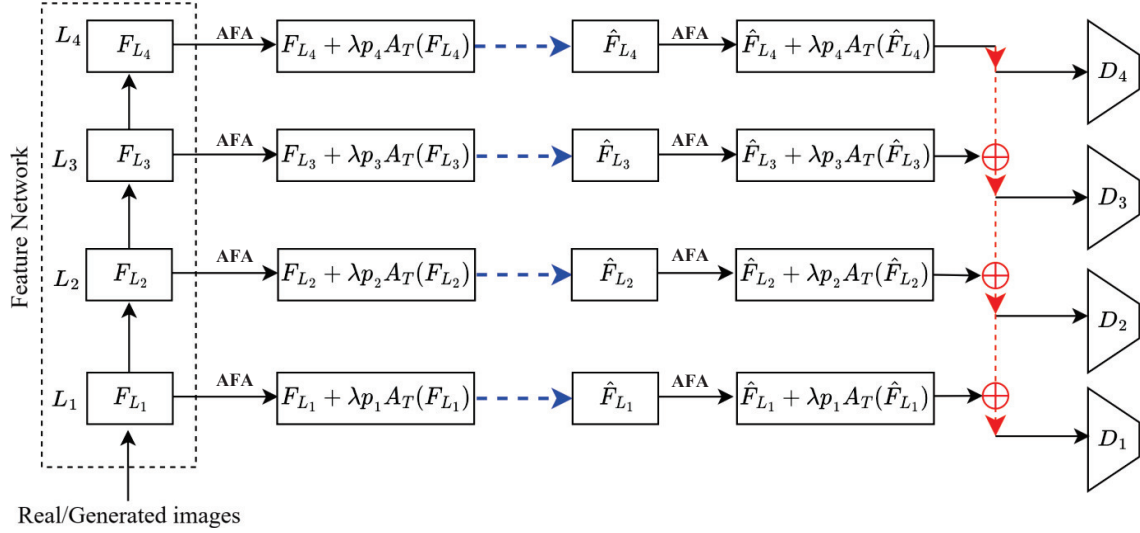


Figure 2. An overview of Adaptive Feature Augmentation (AFA) in the Projected GAN. Following the Projected GAN, we obtain features from four layers L_1 of a pre-trained feature network F at different resolutions, i.e., $L_1 = 64^2$, $L_2 = 32^2$, $L_3 = 16^2$, $L_4 = 8^2$. Cross-Channel Mixing, i.e., CCM (dashed blue arrows), employs 1×1 convolutions with random weights. Cross-Scale Mixing, i.e., CSM (dashed red arrows), adds random 3×3 convolutions and bilinear upsampling, yielding a U-Net [33]. $\{D_l\}$ is a set of independent discriminators operating on different feature projections. The projected features are obtained by applying the pre-trained feature network to real and fake images, and AFA applies the forward diffusion process (A_T) to all projected features before CCM and CSM processes. We introduce the hyperparameters λ and the set of probability $\{p_l\}$ in AFA. Specifically, AFA is applied with probability p_l and skipped with probability $1 - p_l$ in each scale of features, where p_l is controlled by the overfitting of D_l during training. The hyperparameter λ is set at 0.5 to balance the AFA with the original features.

returns +1 for a non-negative input; -1, otherwise. Then, we follow the same step as in ADA [19] and APA [16] for using η_l to adjust p_l . The AFA in each D_l will be applied with the probability p_l or be skipped with the probability $1 - p_l$. In this way, the strength of AFA can be adaptively controlled based on the degree of overfitting, thus alleviating the leaking of augmentation during training.

To sum up, the AFA can be formulated as:

$$\hat{P}_l(x) = P_l(x) + \lambda p_l A_T(P_l(x)), \quad (7)$$

where A_T is the forward diffusion process following Eq.(3), $P_l(x)$ is the original feature produced by the projectors and $\hat{P}_l(x)$ is the obtained augmented features. In forward diffusion process A_T , we select the value of time steps T as 500 and set the forward process variances to constants increasing linearly from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$.

3.2. Theoretical Analysis

Let $\mathbb{P}_{P_l \circ R}$ be the distribution of real data and $\mathbb{P}_{P_l \circ G}$ be the distribution of generated data in projected features distribution. Let λ be the hyperparameter that aims to balance the proposed AFA. For the sample x , $D_l(P_l(x))$ represents the estimated probability of projected features $P_l(x)$ being real or fake. To examine the rationality of AFA, we analyze it in a non-parametric setting, where a model is represented with infinite capacity by exploring its convergence

in the space of probability density functions. Ideally, the estimated probability distribution $\mathbb{P}_{P_l \circ G}$ should perfectly model the distribution $\mathbb{P}_{P_l \circ R}$ without bias if given enough capability and training time.

Since the probability p_l is adaptively adjusted, to facilitate this theoretical analysis, we assume that α_l is the expected strength, which approximates the effect of dynamic adjustment of distribution during the entire training procedure. Since $p_l \in [0, 1]$, we have $0 \leq \alpha < p_{l_{max}} < 1$, where $p_{l_{max}}$ is the maximum probability throughout training. Based on Eq.(2), the loss function of AFA in the Projected GAN is shown as follows:

$$\begin{aligned} \min_G \max_{\{D_l\}} \sum_{l \in \mathcal{L}} & (\mathbb{E}_{x \sim \mathbb{P}_x} [\log D_l(P_l(x) + \lambda \alpha_l A_T(P_l(x)))] \\ & + \mathbb{E}_{z \sim \mathbb{P}_z} [\log(1 - D_l(P_l(G(z)) + \lambda \alpha_l A_T(P_l(G(z)))))]), \end{aligned} \quad (8)$$

where A_T is the forward diffusion process function following Eq.(3). Ideally, for $\forall x$, $A_T(x) \sim N\{0, \sigma^2\}$, it demonstrates that $A_T(x)$ can be directly regarded as Gaussian distribution noise during training. As a result, AFA in the Projected GAN can be theoretically regarded as adding Gaussian distribution noise in feature space during training. In this case, we analyze the convergence of Eq. (8).

For $\forall x$, let $\varepsilon = \lambda \alpha_l A_T(x)$ represent the random Gaussian distribution noise. Then, Eq.(8) can be formulated as:

$$\min_G \max_{\{D_l\}} \sum_{l \in \mathcal{L}} \sum_{\mathbb{P}_\varepsilon \in S} (\mathbb{E}_{x \sim \mathbb{P}_x} [\log D_l(P_l(x) + \varepsilon)] + \mathbb{E}_{z \sim \mathbb{P}_z} [\log(1 - D_l(P_l(G(z)) + \varepsilon))]) \quad (9)$$

According to [15], we introduce a set S of probability density functions in Eq.(9), where $\varepsilon \sim \mathbb{P}_\varepsilon \in S$. Then, based on [15, 36], the optimal discriminator in Eq.(9) can be formulated as:

$$D_l^*(x) = \frac{\sum_{\mathbb{P}_\varepsilon \in S} \mathbb{P}_{P_l \circ R, \varepsilon}}{\sum_{\mathbb{P}_\varepsilon \in S} [\mathbb{P}_{P_l \circ R, \varepsilon} + \mathbb{P}_{P_l \circ G, \varepsilon}]}. \quad (10)$$

Then, according to the [15], the optimization of G under optimal discriminator $D_l^*(x)$ in Eq.(9) can be formulated as:

$$\min_G \mathbf{JSD}(\frac{1}{|S|} \sum_{\mathbb{P}_\varepsilon \in S} \mathbb{P}_{P_l \circ R, \varepsilon} \parallel \frac{1}{|S|} \sum_{\mathbb{P}_\varepsilon \in S} \mathbb{P}_{P_l \circ G, \varepsilon}), \quad (11)$$

where **JSD** is Jensen-Shannon divergence. Following Theorem 1 in [15], we can simplify the Eq.(11) as:

$$\min_G \mathbf{JSD}(\frac{1}{2}(\mathbb{P}_{P_l \circ R} + \mathbb{P}_{P_l \circ R} * \mathbb{P}_\varepsilon) \parallel \frac{1}{2}(\mathbb{P}_{P_l \circ G} + \mathbb{P}_{P_l \circ G} * \mathbb{P}_\varepsilon)), \quad (12)$$

where $\mathbb{P}_\varepsilon$ is a zero-mean Gaussian function produced by AFA with a positive definite covariance Σ . Then, based on the proofs of Theorem 1 in [15], we can conclude that the optimization of Eq.(12) is equal to the optimization between the distributions $P_{P_l \circ G}$ and $P_{P_l \circ R}$, which is the same as in the Projected GAN [36], indicating that the proposed AFA does not influence the convergence.

Next, we prove the proposed AFA can increase the overlapping parts between the supports of $\mathbb{P}_{P_l \circ G}$ and $\mathbb{P}_{P_l \circ R}$, shown in Theorem 1.

Theorem 1. For two supports of the distributions P and Q consisting of overlapping parts, if there exists one feature $P_l(x) \in P \cap Q$, $\forall$ augmentation A , $[P_l(x) + \lambda p_l A(P_l(x))] \in [P + \lambda p_l A(P)] \cap [Q + \lambda p_l A(Q)]$.

Proof. $[P_l(x) \in P \cap Q] \Rightarrow [P_l(x) \in P \text{ and } P_l(x) \in Q] \Rightarrow [\lambda p_l A(P_l(x)) \in \lambda p_l A(P) \text{ and } \lambda p_l A(P_l(x)) \in \lambda p_l A(Q)] \Rightarrow [[P_l(x) + \lambda p_l A(P_l(x))] \in [P + \lambda p_l A(P)] \cap [Q + \lambda p_l A(Q)]]$.

The P and Q in Theorem 1 can be replaced as $\mathbb{P}_{P_l \circ G}$ and $\mathbb{P}_{P_l \circ R}$ for the Projected GAN, and the augmentation A can be replaced as A_T in AFA. Theorem 1 shows that with the augmentation A_T in feature space, if the feature $P_l(x)$ is in the overlapping parts between the supports of the $\mathbb{P}_{P_l \circ G}$ and $\mathbb{P}_{P_l \circ R}$, $P_l(x) + \lambda p_l A_T(P_l(x))$ can still be in the overlapping parts between the supports of the $\mathbb{P}_{P_l \circ G} + \lambda p_l A_T(\mathbb{P}_{P_l \circ G})$ and $\mathbb{P}_{P_l \circ R} + \lambda p_l A_T(\mathbb{P}_{P_l \circ R})$. In this case, due to the AFA in each D_l will be applied with the probability p_l or be skipped with the probability $1 - p_l$, both $P_l(x)$ (under probability $1 - p_l$) and $P_l(x) + \lambda p_l A_T(P_l(x))$ (under probability p_l) are in the overlapping parts between the supports of the $\mathbb{P}_{P_l \circ G} +$

$\lambda p_l A_T(\mathbb{P}_{P_l \circ G})$ and $\mathbb{P}_{P_l \circ R} + \lambda p_l A_T(\mathbb{P}_{P_l \circ R})$. Furthermore, for augmented feature $P_l(x) + \lambda p_l A_T(P_l(x))$, due to forward diffusion process A_T added random Gaussian distribution noise in each step t , one feature $P_l(x)$ can produce a set of augmented feature $\{P_l(x) + \lambda p_l A_T(P_l(x))\}$. These two evidences show that the overlapping parts between the augmentation distribution $\mathbb{P}_{P_l \circ G} + \lambda p_l A_T(\mathbb{P}_{P_l \circ G})$ and $\mathbb{P}_{P_l \circ R} + \lambda p_l A_T(\mathbb{P}_{P_l \circ R})$ consist of more samples, thus providing more overlapping parts, leading to better performance.

3.3. Discussion of the Difference between AFA and Existing studies

Difference between AFA and existing noise injection methods. There are two main differences between AFA and existing noise injection methods. First, existing noise injection methods [1, 15, 34, 38, 42] are all designed in image space. In contrast, the proposed AFA is designed in feature space, which can avoid the leaking of augmentations problem [19] (see section 4.5.1). Second, existing studies [1, 34, 42] point out that directly adding instance noise to the high-dimensional discriminator input can cause the training instability in GANs, which can impact the performance of GANs. On the contrary, applying the forward diffusion process to generate Gaussian-mixture distributed instance noise has shown to be advantageous in Diffusion-GAN [42]. Thus, applying AFA yields better performance than applying random Gaussian distribution noise in feature space (see section 4.5.3).

Difference between AFA and the Diffusion-GAN [42] Similar to the DAs in image space [19, 40, 42, 48], the design of the diffusion in the Diffusion-GAN aims to transform both real and fake samples in image space. It is indeed that Diffusion-Projected GAN (Features) can directly apply their image-space design to feature space to benefit the training of GANs with limited data. However, the design of diffusion in Diffusion-GAN causes the loss of the overlapping parts between the supports of the original real and fake projected feature distributions, thus leading to sub-optimal performance. In contrast, motivated by existing studies [8, 23], the form of the proposed AFA is designed in feature space, allowing it to maintain the overlapping parts between the supports of the original real and fake projected features distributions (section 3.2), thus providing more overlapping parts compared with Diffusion-Projected GAN (Features), leading to better performance (see section 4.5.4).

4. Experiment

4.1. Datasets and Implementation Details

We select FFHQ [21], LSUN-CAT [37] and low-shot datasets for the experiments; all the images are resized to

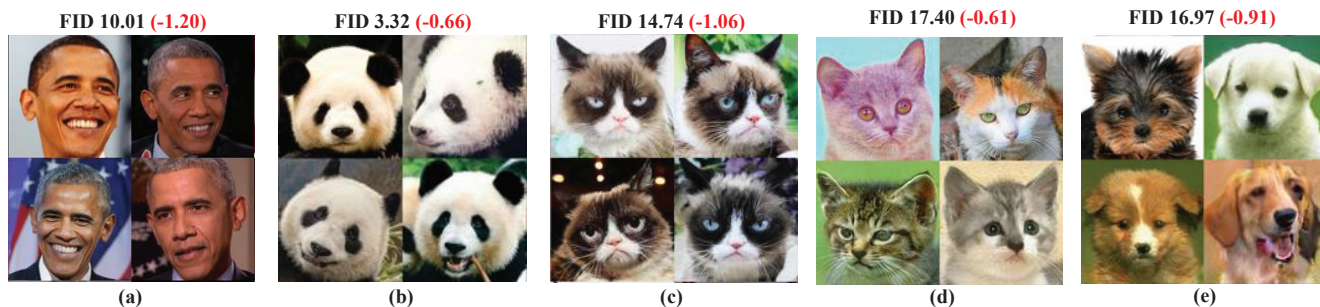


Figure 3. Images generated by Projected GAN + AFA on (a) 100-shot Obama dataset, (b) 100-shot Panda dataset, (c) 100-shot Grumpy Cat dataset, (d) Animal-Face Cat dataset and (e) Animal-Face Dog datasets. The decreasing value of FID in red color demonstrates the improvement of Projected GAN + AFA compared with baseline Projected GAN. *Best viewed in color.*

Method	Backbone	100-shot			Animal-Face	
		Obama	Grumpy Cat	Panda	Cat	Dog
StyleGAN2 [21]	StyleGAN2	65.57	39.92	22.08	51.66	77.96
Diff-Augment [48]	StyleGAN2	46.87	27.08	12.06	42.44	58.85
ADA [19]	StyleGAN2	45.69	26.62	12.90	40.77	56.83
Diffusion-GAN [42]	StyleGAN2	28.55	21.87	8.69	33.18	68.15
InsGen [44]	StyleGAN2	32.42	22.01	9.85	33.01	44.93
FakeCLR [24]	StyleGAN2	26.95	19.56	8.42	26.34	42.02
FastGAN [25]	FastGAN	35.80	25.75	9.70	33.85	52.46
FreGAN [45]	FastGAN	33.39	24.93	8.97	31.05	47.85
Projected GAN [36]	FastGAN	11.21	15.80	3.98	18.01	17.88
Projected GAN + AFA	FastGAN	10.01	14.74	3.32	17.40	16.97

Table 1. FID score (lower is better) on several low-shot datasets (256×256). We follow the setting as in [48]. Massive Augmentation [6] is applied to all of the methods. For a fair comparison, the FIDs are averaged over five runs; all standard deviations are less than 1% relatively. The results of the Projected GAN are run by ourselves based on the official open-source codes <https://github.com/autonomousvision/projected-gan>.

256×256 for a fair comparison. FFHQ dataset contains 70K high-resolution images of human faces. Following FakeCLR [24], we select a subset of 100, 1K, 2K, and 5K for a fair comparison. LSUN-CAT dataset contains 200K high-resolution images of cats, based on the Diff-Augment [48], we select a subset of 1K, 5K, 10K and 30K for the comparison. Low-shot datasets contain five datasets (i.e., Obama, Grumpy Cat, Panda, Animal-Face Cat, and Animal-Face Dog) with 100, 100, 100, 160, and 389 training images, respectively. We select the Projected GAN [36] as our backbone and set the batch size as 64 for experiments; specifically, following the Projected GAN [36], Diff-Augment [48] is applied in image space. The commonly used Fréchet Inception Distance (FID) [13] is applied as the evaluation metric. We select the hyperparameter λ as 0.5 for AFA in all the experiments. In forward diffusion process A_T , we select the value of time steps T as 500 and set the forward process variances to constants increasing linearly from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. We conduct all the experiments on a workstation with four NVIDIA TITAN RTX GPUs. More details

of experiments, i.e., experiments results with the other evaluation metric Inception Score (IS) [35] and the link of the pre-trained model with test code, can be found in supplementary materials, respectively.

4.2. Results on Low-shot Datasets

The results of applying AFA on low-shot datasets are shown in Table 1. Projected GAN + AFA achieves the lowest FID compared with other methods. Specifically, although the baseline Projected GAN has applied the Diff-Augment [48] in image space and achieved impressive results compared with other methods, adding AFA can further reduce the FID score by about 5% to 10%, showing the superiority of AFA. The images generated by Projected GAN + AFA on low-shot datasets are shown in Figure 3.

4.3. Results on FFHQ and LSUN-CAT Datasets

The results of the FFHQ and LSUN-CAT datasets are shown in Table 2. Following the FakeCLR [24], we conduct experiments on the subset of FFHQ 100, FFHQ 1K,

Method	Backbone	FFHQ				LSUN-CAT			
		100	1K	2K	5K	1K	5K	10K	30K
StyleGAN2 [21]	StyleGAN2	179	100.16	54.3	49.68	182.85	34.69	17.93	10.12
ADA [19]	StyleGAN2	85.8	21.29	15.39	10.96	-	-	-	-
Diff-Augment [48]	StyleGAN2	61.91	25.66	24.32	10.45	42.26	16.11	12.07	9.68
APA [16]	StyleGAN2	65	18.89	16.90	8.38	-	-	-	-
InsGen [44]	StyleGAN2	45.75	18.21	11.47	7.83	-	-	-	-
FakeCLR [24]	StyleGAN2	42.56	15.92	9.90	7.25	-	-	-	-
Projected GAN	FastGAN	26.25	11.12	8.25	6.85	19.39	13.57	8.92	8.65
Projected GAN + AFA	FastGAN	23.95	10.55	7.97	6.06	16.78	11.24	8.56	8.11

Table 2. FID score (lower is better) on the 256×256 FFHQ and LSUN-CAT datasets. Following FakeCLR [24] and Diff-Augment [48], we perform experiments on 100, 1K, 2K and 5K training samples on FFHQ dataset and perform experiments on 1K, 5K, 10K and 30K training samples on LSUN-CAT dataset, respectively. Massive Augmentation (MA) is applied in all of the methods. For a fair comparison, the FIDs are averaged over five runs; all standard deviations are less than 1% relatively. The results of the Projected GAN are run by ourselves based on the official open-source codes.

FFHQ 2K and FFHQ 5K. Projected GAN + AFA achieves state-of-the-art performance compared with other methods. Based on Diff-Augment [48], we conduct experiments on the subset of LSUN-CAT 1K, LSUN-CAT 5K, LSUN-CAT 10K and LSUN-CAT 30K. Projected GAN + AFA also achieves state-of-the-art performance compared with other methods. Moreover, although the Projected GAN has applied the Diff-Augment [48] in image space and achieved impressive compared with other methods, adding AFA can further reduce the FID score by about 5% to 10%.

Method	Seconds per 1K images
Projected GAN	12.11
Projected GAN + AFA	12.20

Table 3. The training time on the Animal-Face Dog dataset (256×256) with or without AFA. The results are calculated by averaging over ten times on the four NVIDIA RTX TITAN GPUs with batch size 64. All standard deviations are less than 1% relatively.

4.4. Computational Cost

The results of the training time on the Animal-Face Dog dataset (256×256) with or without AFA using Projected GAN has been demonstrated in Table 3. The proposed AFA only increases the negligible computational cost.

4.5. Ablation Study

To better illustrate the rationality of the proposed AFA, we also conduct a series of ablation studies on the proposed AFA as follows.

4.5.1 AFA in feature space v.s. AFA in image space.

To demonstrate the reasonable design of AFA in feature space rather than in image space, we conduct an ablation



Figure 4. Images generated by Projected GAN + AFA on 100-shot Obama dataset. (a) Applying AFA in image space and (b) AFA in feature space. Best viewed in color.

Method	FID
AFA in image space	13.65
AFA in feature space	10.01

Table 4. FID score (lower is better) on the 100-shot Obama dataset (256×256). Massive Augmentation [6] is applied to all of the methods. For a fair comparison, the FIDs are averaged over three runs; all standard deviations are less than 1% relatively. The Projected GAN are selected as backbones for this experiment.

study applying the AFA in image space of the Projected GAN. Specifically, for real or fake images, we apply the forward diffusion process to augment themselves in image space. For better visualization results, we select the 100-shot Obama dataset for the experiments, and demonstrate results in Table 4 and Figure 4. As shown in Figure 4, it is clear that applying AFA in image space can still produce noise-based images, which is caused by the leaking of augmentations problem [19] during training. In contrast, applying AFA in feature space can avoid the leaking of augmentations during training, thus resulting in better performance.

Method	100-shot			Animal-Face	
	Obama	Grumpy Cat	Panda	Cat	Dog
Diffusion-Projected GAN (Features) [42]	10.54	15.13	3.39	17.86	17.22
Projected GAN + AFA	10.01	14.74	3.31	17.39	16.97

Table 5. FID score (lower is better) on several low-shot datasets (256×256). We follow the setting as in [48]. Massive Augmentation [6] is applied to all of the methods. For a fair comparison, the FIDs are averaged over three runs; all standard deviations are less than 1% relatively.

Method	FID
Projected GAN + AFA without adaptive process	17.34
Projected GAN + AFA	16.97

Table 6. FID score (lower is better) on the Animal-Face Dog dataset (256×256). We follow the setting as in [48]. Massive Augmentation [6] is applied to all of the methods. For a fair comparison, the FIDs are averaged over three runs; all standard deviations are less than 1% relatively.

Method	FID
Projected GAN + Random Gaussian noise	17.57
Projected GAN + AFA	16.97

Table 7. FID score (lower is better) on Animal-Face Dog dataset (256×256). We follow the setting as in [48]. Massive Augmentation [6] is applied to all of the methods. For a fair comparison, the FIDs are averaged over three runs; all standard deviations are less than 1% relatively.

4.5.2 AFA v.s. AFA without adaptive process.

To demonstrate the designing of the adaptive process in AFA is reasonable, we conduct an ablation study comparing applying AFA without the adaptive process, and the results are shown in Table 6. The designing of the adaptive process in AFA can indeed improve performance.

4.5.3 AFA v.s. Directly adding noise in feature space.

Following the analysis in section 3.3, we conduct the ablation study to compare applying AFA and directly adding random Gaussian distribution noise in feature space. The results are shown in Table 7. AFA can achieve better performance compared with directly adding random Gaussian distribution noise in feature space.

4.5.4 AFA v.s. Transforming both real and fake features by diffusion.

According to [19, 40, 42, 48], the common way of DA in training GANs with limited data is to transform both real and fake samples. Therefore, we also conducted an ablation

study comparing the proposed AFA with one commonly used transforming both real and fake features by diffusion method, i.e., Diffusion-Projected GAN (Features) [42], and the results are shown in Table 5. AFA can achieve better performance compared with Diffusion-Projected GAN (Features).

4.5.5 Effectiveness of the hyperparameter λ .

To demonstrate how to select hyperparameter λ in the proposed AFA, we also conduct an ablation study on selecting different hyperparameters λ in AFA, and the results are shown in Table 8. These results drive us to select λ as 0.5 for all of the experiments.

λ	0.25	0.50	0.75	1.00
FID	17.04	16.97	17.12	17.26

Table 8. FID score (lower is better) on the Animal-Face Dog dataset (256×256). We follow the setting as in [48]. Massive Augmentation [6] is applied to all of the methods. For a fair comparison, the FIDs are averaged over three runs; all standard deviations are less than 1% relatively.

5. Conclusion

In this paper, to improve the training of the Projected GAN with limited data, we propose a simple yet effective feature augmentation for the Projected GAN, called Adaptive Feature Augmentation (AFA), with a negligible computational cost increase. Specifically, we apply the forward diffusion process to the projected features obtained from both real and fake images to augment projected features themselves, which can effectively provide more overlapping parts between the supports of real and fake projected features distributions during training, thus leading to better performance. Extensive experiments on several low-shot datasets demonstrate that AFA can effectively improve the training of the Projected GAN and achieve state-of-the-art results compared with other methods.

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